A multidimensional image quality measure using Singular Value Decomposition

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ABSTRACT

The important criteria used in subjective evaluation of distorted images include the amount of distortion, the type of distortion, and the distribution of error. An ideal image quality measure should therefore be able to mimic the human observer. We present a new image quality measure that can be used as a multidimensional or a scalar measure to predict the distortion introduced by a wide range of noise sources. Based on the Singular Value Decomposition, it reliably measures the distortion not only within a distortion type at different distortion levels but also across different distortion types. The measure was applied to Lena using six types of distortion (JPEG, JPEG 2000, Gaussian blur, Gaussian noise, sharpening and DC-shifting), each with five distortion levels.

Keywords: image quality, objective measures, subjective evaluation, mean squared error, local error measurement, human visual system, visual threshold, singular value decomposition.

1. INTRODUCTION

Measurement of image quality is a challenging problem in many image processing fields from image compression to printing. In the past 30 years, a vast literature has appeared with many approaches attempting to provide a solution [1]. The image quality measures in the literature can be classified into two groups: Subjective and objective [2]. Subjective evaluation is cumbersome as the human observers can be influenced by several critical factors including the environmental conditions, motivation and mood. The objective measures include bivariate measures such as the Mean Squared Error (MSE) or L_p-norm [3,4,5,6], measures mimicking the human visual system (HVS) [4,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35], and graphical measures [33,36,37,38,39,40,41,42,43]. Furthermore, several works present comparative evaluations of a number of selected measures for image compression [14,16,28,31,40,44], and for noise and blur [13,45]. The most common objective evaluation tool, the MSE, is very unreliable, resulting in poor correlation with the HVS. In spite of their complicated algorithms, the HVS-based objective measures do not appear to be superior to the simple pixel-based measures like the MSE, Peak Signal-to-Noise Ratio (PSNR) or Root Mean Squared Error (RMSE).

A number of researchers point to the disadvantages of the measures that incorporate an HVS model. Fuhrmann et al [45] discourage the use of metrics based on the spatial frequency properties of the HVS as they require precise knowledge of the viewing conditions. Franti [30] argues that the distortion measure should be independent of the factors such as the compression method used, basic image processing operations, and the viewing distance. According to Wang and Bovik [34], although the viewing conditions play an important role in human perception of image quality, they are not fixed in most cases, and the specific data is generally unavailable to the image analysis system.

An ideal image quality measure should be able to describe

1. the amount of distortion,
2. the type of distortion, and
3. the distribution of error.

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Such a measure is expected to provide accurate predictions of quality not only at distortion ranges near the visual threshold but also when distortions are significantly above the visual threshold. Undoubtedly, there is a need for an objective measure that provides more information than a single numerical value. Assessment of image quality is an open problem today.

In their 1982 paper [11], Lukas and Budrikis talk about a possible improvement in quality prediction if local rather than global averaging procedures are used. They believe local error measures are particularly pertinent in the case of coding schemes that introduce distortion that is very localized in nature. Westen et al [24] present a perceptual error measure (PEM) based on a multiple channel HVS for use in digital image compression. PEM combines the responses in different frequency bands, orientations and positions. Combination of the responses at each position results in an image with values that represent a local visibility of distortions. In coding applications, the authors believe, such a local measure of image quality is probably more useful than a global one. Eude and Mayache [31] compare four metrics to evaluate the quality of JPEG-compressed images. They conclude by stating that as these metrics do not take into account typical artifacts of other compression methods, a multi-dimensional measure, with each dimension being related to an artifact, would be an attractive solution.

A recent paper [34] presents a new numerical measure, called a universal image quality index, which is defined as

$$Q = \frac{4\sigma_{xy}\overline{x}\overline{y}}{(\sigma_x^2 + \sigma_y^2)[(\overline{x})^2 + (\overline{y})^2]},$$

where $x_i, y_i, i = 1,...n$, represent the original and distorted signals, respectively, $\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$, $\overline{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$, $\sigma_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$, $\sigma_y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \overline{y})^2$, and $\sigma_{xy}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})$.

The dynamic range of Q is [-1,1], with the best value achieved when $y_i = x_i$, i=1,2,…,n. As described in the paper, this index models any distortion as a combination of three different factors: loss of correlation, mean distortion and variance distortion. It is applied to Lena using a sliding window approach with a window size of 8x8. The index is computed for each window, leading to a quality map of the image. The overall quality index is the average of the quality map. The test image is corrupted by seven types of distortions: Impulsive salt-pepper noise, additive Gaussian noise, multiplicative speckle noise, mean shift, contrast stretching, blurring, and JPEG compression. The results are compared with MSE and subjective evaluation.

In this paper, we present a new measure that can express the quality of distorted images either graphically (as a two-dimensional measure) or numerically (as a scalar measure) both near and above the visual threshold. Based on the Singular Value Decomposition (SVD), it consistently measures the distortion both across different distortion types and within a given distortion type at different distortion levels.

2. A NEW SVD-BASED MEASURE

Every real matrix A can be decomposed into a product of 3 matrices $A = USV^T$, where U and V are orthogonal matrices, $U^TU = I$, $V^TV = I$ and $S = diag\ (s_1, s_2, ...)$. The diagonal entries of S are called the singular values of A, the columns of U are called the left singular vectors of A, and the columns of V are called the right singular vectors of A. This decomposition is known as the Singular Value Decomposition (SVD) of A [46]. It is one of the most useful tools of linear algebra with several applications to multimedia including image compression [47] and watermarking [48,49,50].

We will restrict our description to gray scale images although generalization to color images is possible. Every gray scale image can be considered to be a matrix with an integer number corresponding to each pixel. If the SVD is applied to the full images, we obtain a global measure whereas if a smaller block (e.g., 8x8) is used, we compute the local error in that block. An alternative for the global measure is to obtain the local errors in smaller blocks and average them in a certain way.

2.1 Graphical measure

The proposed graphical measure is a bivariate measure that computes the distance between the singular values of the original image block and the singular values of the distorted image block:

$$D_i = Sqrt[\sum_{i=1}^{n}(s_i - \hat{s}_i)^2] \ ,$$

where s_i are the singular values of the original block, $\hat{s}_i$ are the singular values of the distorted block, and n is the block size. If the image size is k, we have (k/n) x (k/n) blocks. The set of distances, when displayed in a graph, represents a "distortion map."

We applied the measure to Lena, one of the most widely used test images. The range of its pixel values is [24, 254]. The distortion types, the distortion levels and the associated parameters are given in Table 1. The block size used in our experiments is 8 as this is a common block size in JPEG compression and other image processes.

Table 1. Distortion types and levels

Type \ Level	Level 1	Level 2	Level 3	Level 4	Level 5
JPEG	10:1	20:1	30:1	40:1	50:1
JPEG2000	10:1	20:1	30:1	40:1	50:1
Gaussian blur	1	2	3	4	5
Gaussian noise	3	6	9	12	15
Sharpening	10	20	30	40	50
DC-shifting	4	8	12	16	20

We will begin with the observation that a black image of size n with pixels values equal to 0 have n singular values that are equal to 0, and a white image of size n with pixels values equal to 255 have one singular value that is equal to $nx255$, and n-1 singular values that are equal to 0.

For each distorted image, Figure 1 shows the result of our graphical measure, expressed as a distortion map, that provides the amount of distortion, the type of distortion, and the distribution of error.

- JPEG: As the distortion level is increased, the image becomes blocky (which is the major artifact for Discrete Cosine Transform (DCT)-based JPEG compression). This artifact becomes visible in the graphs starting from compression ratio 30:1, especially on Lena's shoulder and the wall.
- JPEG2000: As this recent compression standard is based on the wavelet transform, the images become blurry along the edges and in high frequency areas. As we increase the compression ratio, the graphs display how the image loses its fidelity. When compared with JPEG, this technology is superior especially at higher compression ratios.
- Gaussian blur: This type of distortion substantially affects the edges and high frequency areas, resulting in seriously blurred images. As the radius of blurring is increased, we see high peaks in the graphs along the edges and high frequency areas.
- Gaussian noise: The effect is a uniformly distributed noise across the image which is depicted in the graphs as the amount of noise goes up. The noise is visible in high frequency, low frequency and textured areas.
- Sharpening: This type of filter makes the textured and high frequency areas sharper and crispier. The graphs show the distortion in the affected areas. In contrast, sharpening does not introduce noticeable noise in the low frequency areas.
- DC-shifting: If a constant value is added to all the pixel values, the image becomes uniformly lighter, and if a constant value is subtracted from all the pixel values, the image becomes uniformly darker. Because of the range of pixel values of Lena (24-254), we subtracted values that resulted in darker areas along the edges with a sharp contrast. As smaller pixel values led to smaller SVs, our measure computed smaller differences along those edges, resulting in "grooves" in the graphs. In the other areas, the distribution of distortion is uniform.

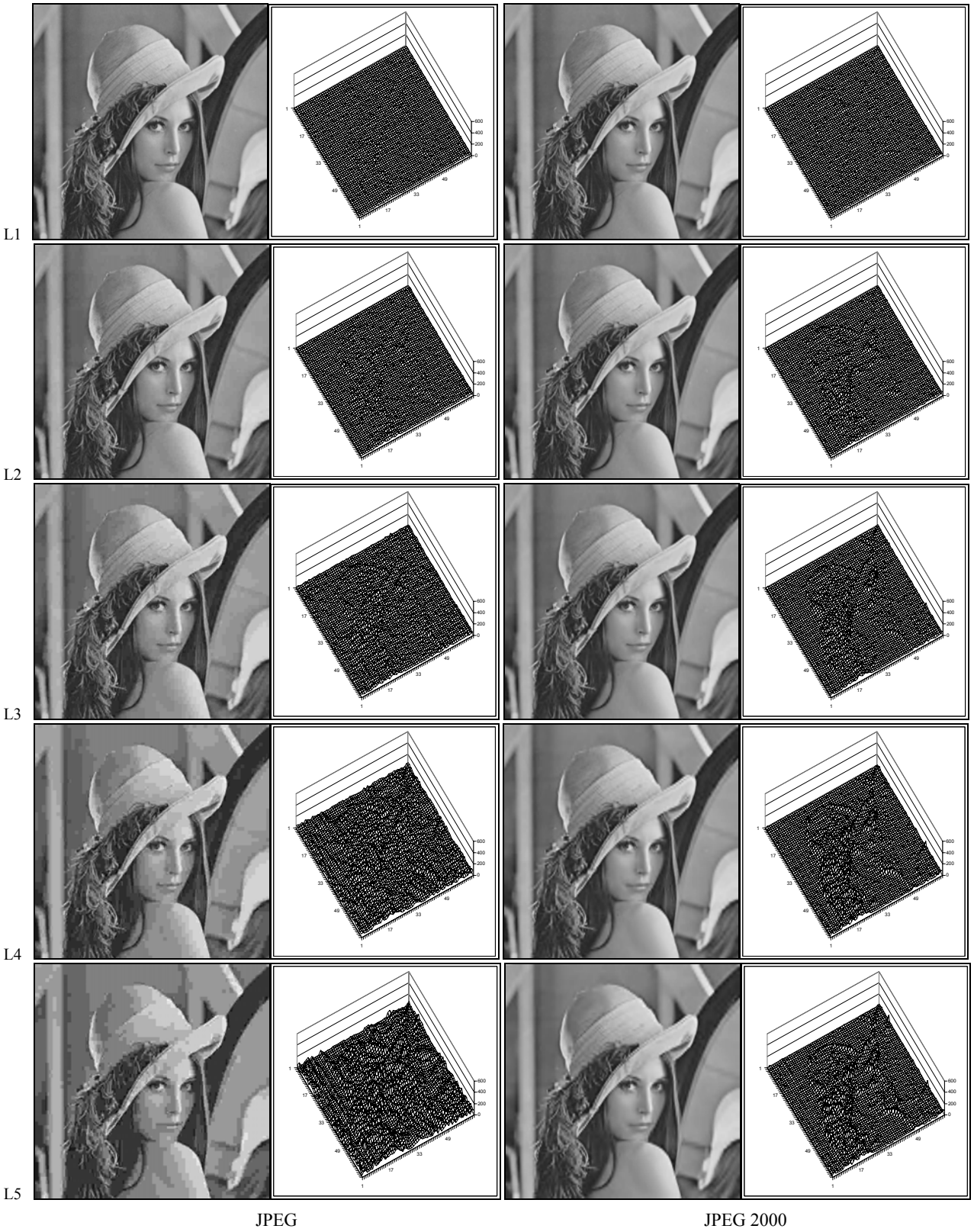
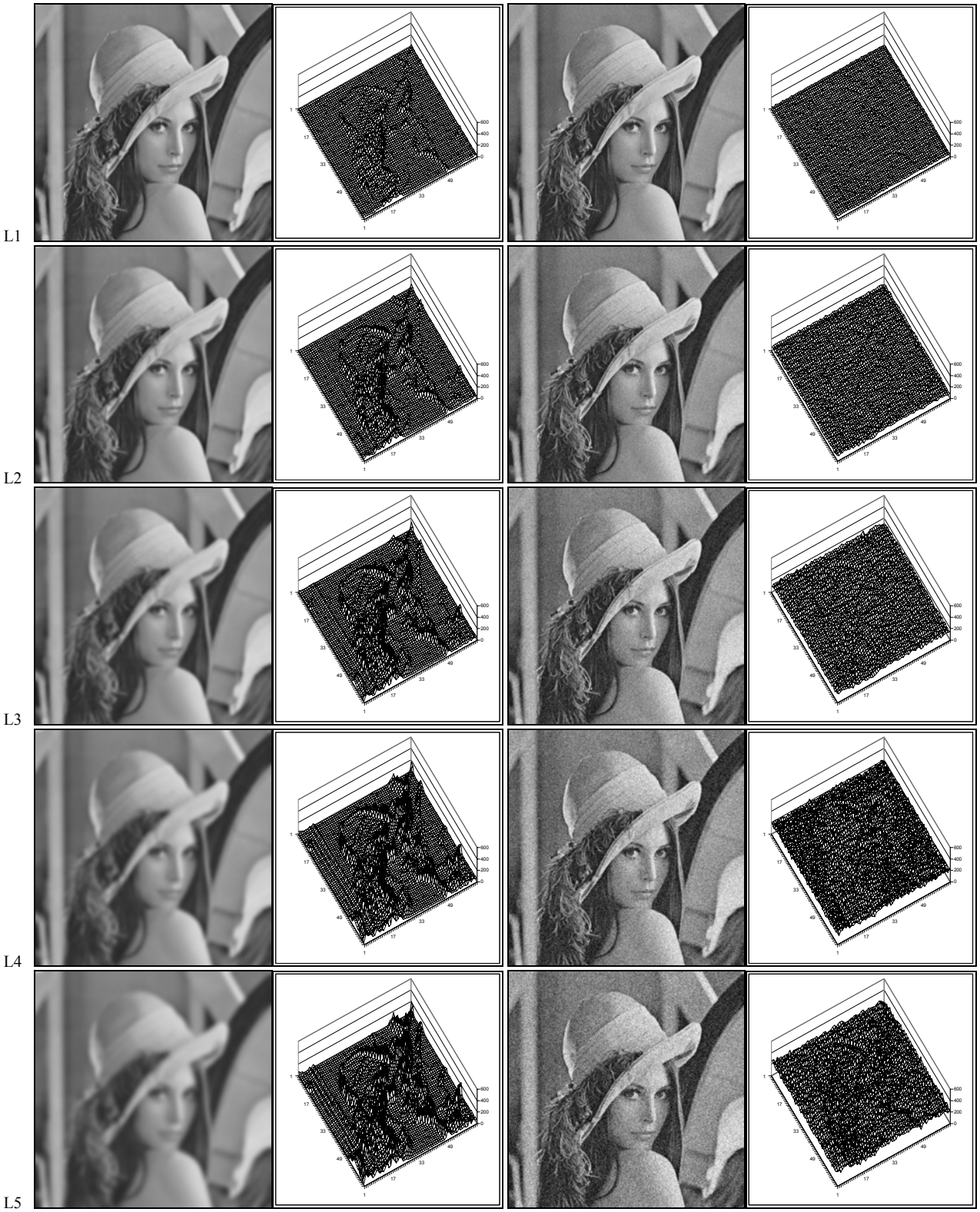
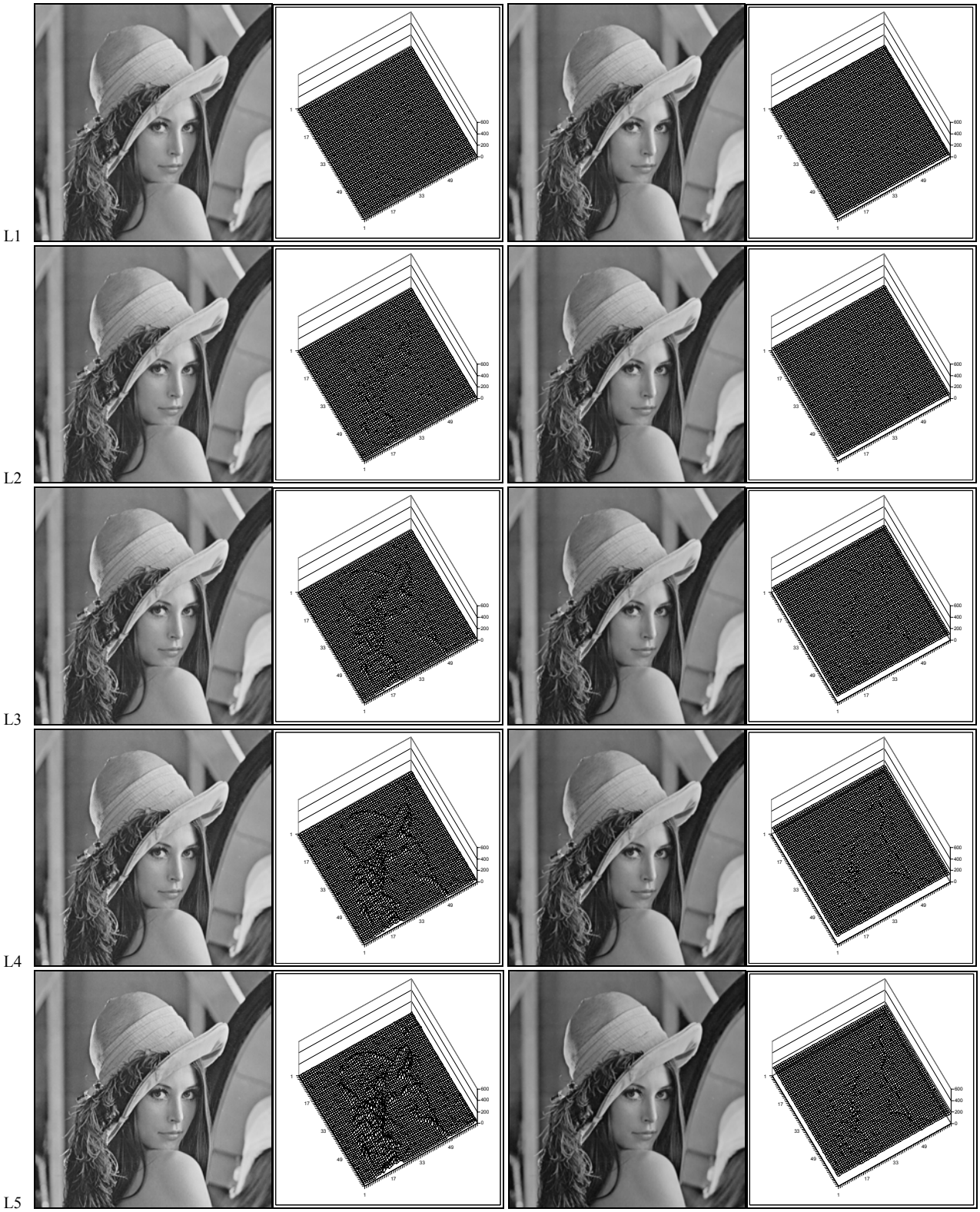


Figure 1. Distorted images and corresponding distortion maps (L*i*, *i*=1,..,5 denote the distortion levels)



Gaussian blur Gaussian noise

Figure 1. Continued



Sharpening DC-shifting

Figure 1. Continued

2.2 Numerical measure

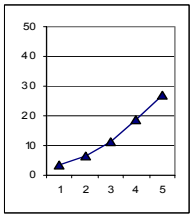
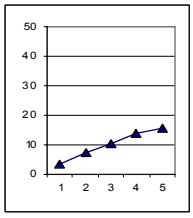
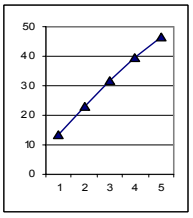
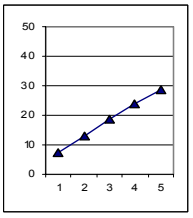
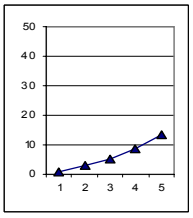
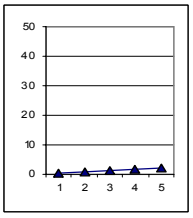
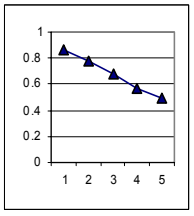
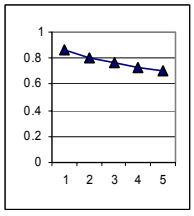
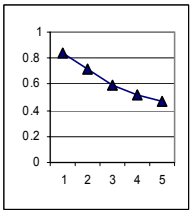
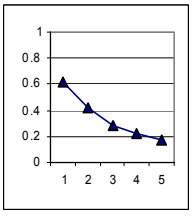
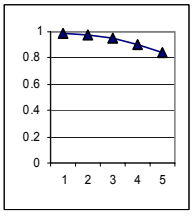
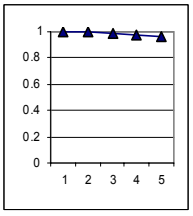
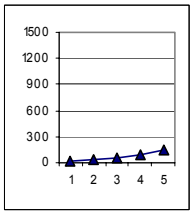
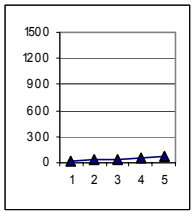
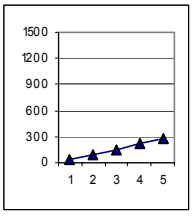
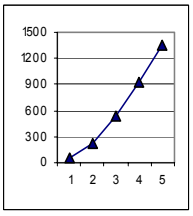
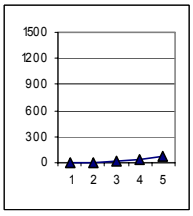
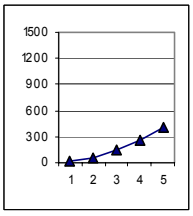
The numerical measure is derived from the graphical measure. It computes a global error expressed as a single numerical value based on the distortion type:

$$M_{SVD} = \frac{\sum_{i=0}^{(k/n)x(k/n)} | D_i - D_{mid} |}{(k/n)x(k/n)},$$

where D_{mid} represents the mid point of the sorted D_i s, k is the image size, and n is the block size.

In Table 2, we compare M_{SVD} for all 30 distorted images with two other measures – the universal image quality index Q and the MSE.

Table 2. Performance of M_{SVD}, Q and MSE.

	JPEG	JPEG2000	G. Blur	G. Noise	Sharpening	DC-Shifting
M_{SVD}						
Q						
MSE						

High quality print-outs of the distorted images were subjectively evaluated by ten observers. In this experiment, the observers were chosen among the undergraduate/graduate students and professors from the Department of Computer and Information Science at Brooklyn College. Five of the observers were very familiar with image processing, and the others only had computer science background. They were asked to rank the images in two ways: Within a given distortion type (i.e, ranking of the 5 distorted images) and across six distortion types (i.e., ranking of the 6 distorted images for each distortion level). As the proposed measure is not HVS-based, no viewing distance was imposed on the observers in the experiment. There were 50 grades in the scale; 1 was assigned to the best image and 50 was assigned to the worst image. The correlation between the three objective measures (i.e., M_{SVD}, Q and MSE) and subjective evaluation is given in Table 3.

We observe that the correlation coefficients for M_{SVD} are higher than those for Q and MSE within each distortion type and across the distortion types. The performance difference is more pronounced in Part (b) that represents a more challenging measurement problem. As expected, the MSE produces the worst correlation.

Table 3. (a) Correlation coefficient within each distortion type

	JPEG	JPEG2000	G. Blur	G. Noise	Sharpening	DC-Shifting
M-SVD	0.996	0.982	0.999	0.995	0.991	0.992
Q	-0.995	-9.980	-0.997	-0.932	-0.970	-0.962
MSE	0.986	0.992	0.995	0.996	0.936	0.971

(b) Correlation coefficient across six distortion types

	Level 1	Level 2	Level 3	Level 4	Level 5
M-SVD	0.934	0.985	0.950	0.941	0.935
Q	-0.783	-0.720	-0.778	-0.830	-0.855
MSE	0.858	0.580	0.400	0.310	0.265

3. CONCLUSIONS

We presented a new image quality measure that can be used graphically as a two-dimensional tool or numerically as a scalar metric. Our observations regarding the proposed measure are as follows:

- The numerical measure is expressed as a Minkowski metric $(\sum_k |s_k - \hat{s}_k|^{\beta})^{1/\beta}$, where β=2, and the image components s_k and $\hat{s}_k$ represent the SVD singular values of the original and distorted images, respectively. It computes a global estimate of the distortion in the image. The subjective evaluation shows that its performance is better than that of the universal quality index Q for all distortion types and distortion levels.

- The graphical measure consistently displays the type and amount of distortion as well as the distribution of error in all the images. In the experiments, we used a wide range of distortion types including compression, blur, noise, sharpening and shifting. Some other measures have limited scope as they focus on a particular technology such as image compression [24,25,26,29,30,31,45].

- Neither the graphical measure nor the numerical measure requires a simplified model of the HVS, necessitating undue computations. Hence, they do not have any assumptions concerning the viewing distance, or the distortion type.

- In some global metrics, the statistics obtained about the impairments in a distorted image are combined in a weighted sum to represent the average error. This weighting is usually problematic as no systematic method is known to determine the weights. Researchers have resorted to simple addition of errors [25], Minkowski metric with different values of β [28], linear combination of distortion factors [22,29,30], and nonlinear response in different frequency bands, orientations and positions [24]. These efforts may result in different weights for different distortions or image types. The M_{SVD}, however, does not require any analysis to compute a weighted sum in predicting the overall error.

- The SVD is of order $O(n^3)$, which makes it slow for large size images. If the image is segmented into smaller blocks, and the SVD is applied to each block, the total processing time is much lower. As we use 8x8 blocks, computational requirements are fairly reasonable.

- The smallest parameters in Table 1 correspond to distortions that are barely visible, and the larger parameters to distortions where the observer preferences start deviating. Hence, the measure is able to reliably predict visual quality not only near the visual threshold but also well above the visual threshold.

To the best of our knowledge, such a generalized metric that can be used for local and global measurements does not exist in the current literature.

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